# Value and Momentum Everywhere

Asness, Moskowitz and Pedersen (2013), *Journal of Finance*

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# The research question

What do value and momentum have in common across markets?

- Most prior evidence focuses on U.S. individual stocks.
- Value and momentum are often examined separately.
- This paper studies both jointly across eight markets.

| **Four individual-stock markets** |
| --- |
| United States, United Kingdom, continental Europe, Japan |
| **Four markets beyond individual stocks** |
| Equity indices, currencies, government bonds, commodities |

# The main findings

1. **Returns:** value and momentum have positive sample premia across markets, with important weak cases.

2. **Comovement:** value moves with value elsewhere, momentum with momentum elsewhere; the two styles move oppositely.

3. **Explanation:** global style factors capture substantial return variation. Funding liquidity explains only part of the evidence.

Joint evidence creates a harder test for asset pricing theories.

# Data and the investable universe

| Market | Number (minimum) | Data start | Main sources |
|---|---|---|---|
| *Individual stocks: average number of firms* | | | |
| U.S. | 724 (354) | 1972.01 | CRSP, Compustat |
| U.K. | 147 (76) | 1972.01 | Datastream, Worldscope |
| Europe | 290 (96) | 1974.01 | Datastream, Worldscope |
| Japan | 471 (148) | 1974.01 | Datastream, Worldscope |
| *Other markets: total coverage, not average number* | | | |
| Equity indices | 18 (8) | 1978.01 | MSCI, Bloomberg |
| Currencies | 10 (7) | 1979.01 | Datastream, LIBOR |
| Government bonds | 10 (5) | 1982.01 | Bloomberg, Morgan Markets |
| Commodities | 27 (10) | 1972.01 | Futures exchanges |

Monthly data, ending July 2011. Strategy samples can start later.

Stocks cover the top 90% of market capitalization: approximately the largest 17%, 13%, 20%, and 26% of firms, respectively.

Large, liquid securities make the comparison more implementable and the estimated premia conservative in this sample.

# Value and momentum signals

**Cross-sectional momentum:** rank assets by their cumulative return over months 2–12 before formation.

| **Value signal** | **Higher signal means relatively cheaper** |
|---|---|
| Individual stocks | BE/ME: book equity lagged 6 months; recent market equity |
| Equity indices | Previous month's BE/ME of the country's MSCI index |
| Commodities | Negative log five-year spot-price change |
| Currencies | Five-year exchange-rate change adjusted for relative CPI, oriented toward cheapness (PPP measure) |
| Government bonds | Current 10-year yield minus its level five years earlier |

For commodities and currencies, the historical spot-price reference averages months 54–66 before formation.

In **stocks**, five-year reversal and BE/ME value factors correlate 0.83 in the U.S. and 0.86 across the four stock markets.

# Test portfolios and traded factors

**Test assets:** 3 groups $\times$ 2 signals $\times$ 8 markets $=$ 48.

Low / middle / high; stocks value-weighted, other markets equal-weighted.

**Factors:** weights use demeaned signal ranks across the full universe.

$$w_{it}^{S} = c_t \left( \text{rank}(S_{it}) - \frac{1}{N} \sum_{j} \text{rank}(S_{jt}) \right)$$

| Illustrative rank | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| Demeaned rank | $-1.5$ | $-0.5$ | 0.5 | 1.5 |
| Weight ($c_t = 0.5$) | $-0.75$ | $-0.25$ | 0.25 | 0.75 |

$c_t$ scales the factor to \$1 long and \$1 short: **dollar-neutral**, not necessarily beta-neutral. A middle rank can have zero weight.

$$r_t^{\text{COMBO}} = 0.5 r_t^{\text{VALUE}} + 0.5 r_t^{\text{MOM}}$$

# Positive sample returns, with weak cases

| Annualized Sharpe ratio | Value | Momentum | 50/50 |
|---|---|---|---|
| U.S. stocks | 0.26 | 0.45 | 0.86 |
| U.K. stocks | 0.38 | 0.48 | 1.07 |
| Europe stocks | 0.54 | 0.75 | 1.20 |
| Japan stocks | 0.77 | 0.13 | 0.88 |
| Country indices | 0.60 | 0.63 | 1.00 |
| Currencies | 0.44 | 0.32 | 0.69 |
| Government bonds | 0.07 | 0.17 | 0.20 |
| Commodities | 0.31 | 0.51 | 0.77 |

All entries use **signal-weighted factors**, before transaction costs.

**Japan:** Cultural biases are a possible explanation for weak momentum (Chui et al., 2010), not a cause established here. Strong value also needs explaining.

**Bonds:** Five-year yield change may be a weak value proxy; real yield and term spread work better. This does not explain weak bond momentum.

# Combining the styles improves the global portfolio

Annualized Sharpe ratios across all eight markets

| | |
|---|---|
| Value | 0.72 |
| Momentum | 0.74 |
| 50/50 combination | 1.59 |

$$w_m = \frac{1/\sigma_m^{\text{benchmark}}}{\sum_{k=1}^{8} 1/\sigma_k^{\text{benchmark}}}, \qquad \sum_{m=1}^{8} w_m = 1.$$

$\sigma_m^{\text{benchmark}}$: sample volatility of market $m$'s benchmark returns. The same market weight applies to its portfolios and factors.

Cross-market comovement limits diversification within each style. Negative value–momentum correlation creates a further diversification benefit.

The combined Sharpe ratio raises the hurdle for risk-based explanations.

# The correlation structure across markets

| | Stock value | Nonstock value | Stock momentum | Nonstock momentum |
|---|---|---|---|---|
| Stock value | 0.68* | 0.15* | −0.53* | −0.26* |
| Nonstock value | | 0.07 | −0.16* | −0.13* |
| Stock momentum | | | 0.65* | 0.37* |
| Nonstock momentum | | | | 0.21* |

**Off diagonal:** correlations between volatility-weighted group return series.

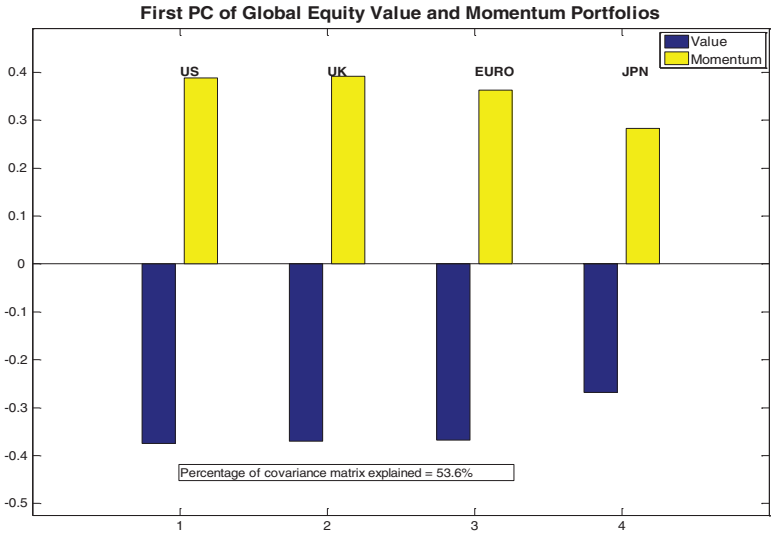
**Diagonal:** average correlation of each market with the aggregate of the other markets in its group. It is not self-correlation.

Quarterly factor returns. * Joint F-test significant at 5%, as in the original table.

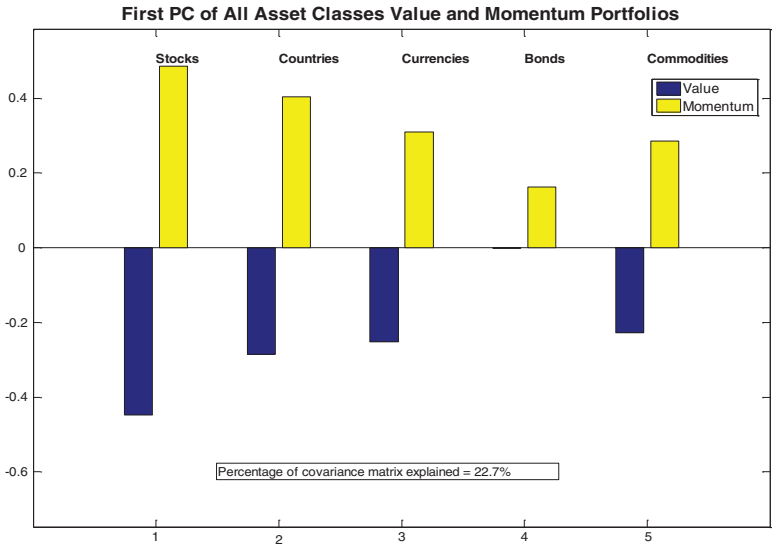
The aggregate sign pattern is consistent; nonstock value–value is not significant. Individual market pairs are less uniform.

# A common component with opposite style loadings

Four stock markets




Five asset groups




Blue: value. Yellow: momentum. First component explains approximately 54% and 23% of total variance, respectively.

The right panel combines the four stock markets into one stock group. Value and momentum have opposite-signed eigenvector weights; bond value is close to zero.

This is evidence of common return variation, not identification of an economic cause.

# Global market and other-market styles: a partial explanation

For each of the 48 test portfolios, regress its excess return on:

1. The global market factor (MSCI World excess return).
2. Value factors aggregated across all other markets.
3. Momentum factors aggregated across all other markets.

| Result | Interpretation |
|---|---|
| $R^2_{\mathrm{CS}} = 0.55$ | Partial fit to differences in average returns across the 48 portfolios. |
| Average $\lvert\alpha\rvert \approx 23$ bp/month | Average returns still deviate from model-implied returns; this is not a significance test. |

Only the **value and momentum factors** exclude the target market; the global market factor does not.

# A global three-factor model

$$r^e_{p,t} = \alpha_p + \beta_p \mathrm{MKT}_t + v_p \mathrm{VAL}_t^{\text{everywhere}} + m_p \mathrm{MOM}_t^{\text{everywhere}} + \epsilon_{p,t}$$

Same factors for every asset; global styles now include all eight markets.

| 48 global portfolios | $R^2_{\mathrm{CS}}$ | Avg. $\lvert\alpha\rvert$ bp/month | GRS $F$ | $p$ |
|---|---|---|---|---|
| Global CAPM | 0.52 | 35 | 6.02 | $< 0.001$ |
| U.S. FF 4-factor | 0.55 | 35 | 6.70 | $< 0.001$ |
| U.S. FF 6-factor | 0.62 | 35 | 7.11 | $< 0.001$ |
| Global 3-factor | 0.72 | 18 | 2.66 | $< 0.001$ |

Better fit, but significant pricing errors remain.

GRS tests whether all alphas are jointly zero. FF 4 = U.S. market, SMB, HML, UMD; FF 6 adds TERM and DEF.

# Funding liquidity exposures have opposite signs

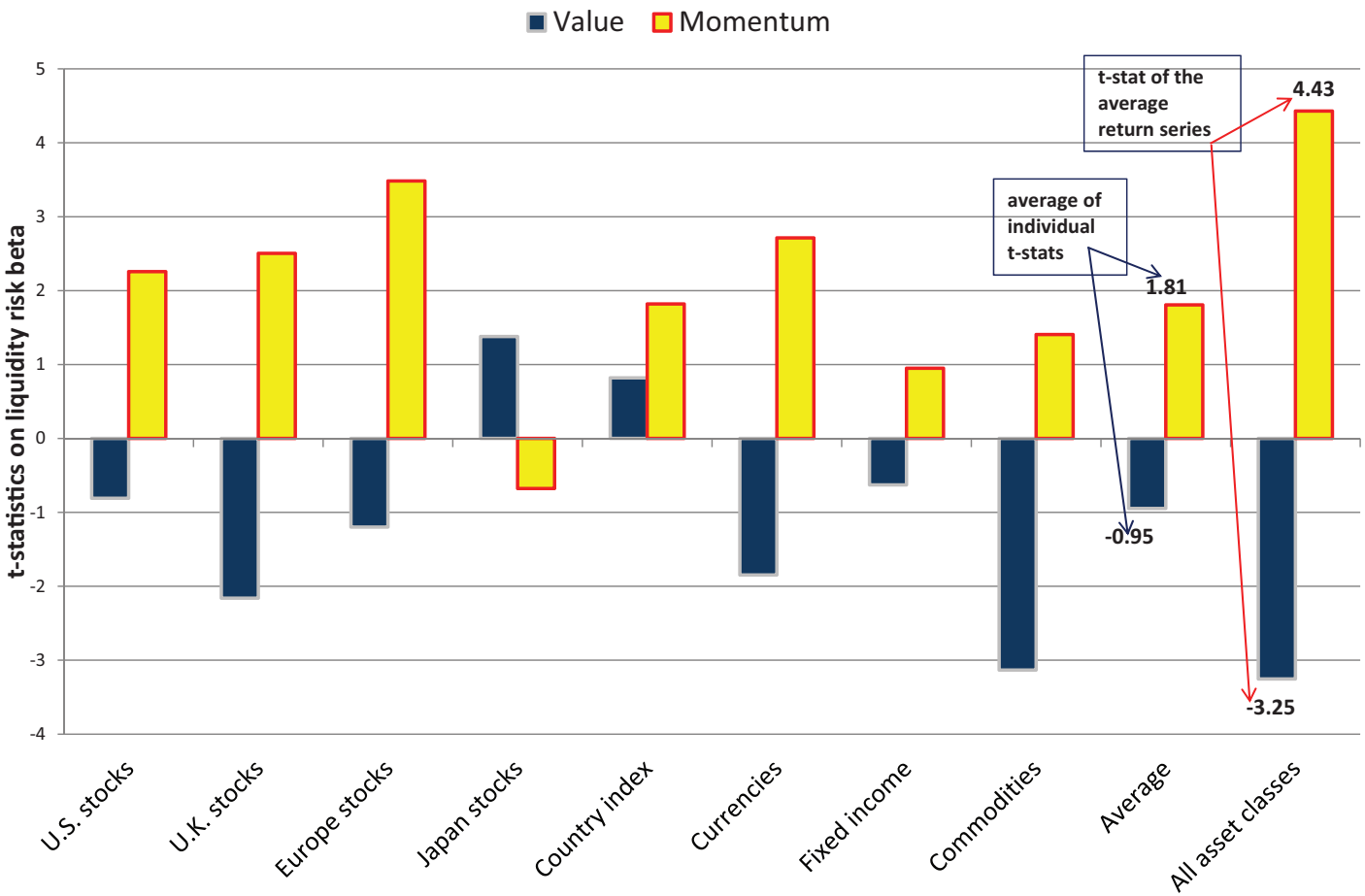
Dependent variables: global strategy returns. Regressors: AR(2) liquidity shocks, with macroeconomic controls. **Positive shock = more liquidity.**

| Global liquidity index | Value | Momentum | 50/50 |
|---|---|---|---|
| Funding PC | −0.0094 | 0.0112 | 0.0013 |
| | (−4.74) | (3.58) | (0.58) |
| Market PC | −0.0009 | 0.0016 | 0.0012 |
| | (−0.74) | (1.21) | (1.00) |
| All PC | −0.0079 | 0.0093 | 0.0016 |
| | (−3.25) | (4.43) | (0.82) |

Coefficients, with $t$-statistics in parentheses. PC = first principal-component index.

Funding exposure is significant for both styles. Market-PC and combination exposures are not statistically significant here.

# Pooling markets reveals the common exposure




Some individual markets are significant; the pattern is uneven. Pooling returns reduces market-specific noise and strengthens the common signal.

# A possible mechanism: funding constraints and crowding

**Interpretation proposed by the authors, not a direct test of holdings.**

| **Momentum** | **Value** |
|---|---|
| Recent winners may attract investors and become crowded. | Contrarian positions may be less crowded. |
| Funding stress may force selling in popular positions. | Less crowded positions may experience less liquidation pressure. |

This offers an intuition for different liquidity exposures. It does not establish the observed magnitudes or fully explain long–short strategy returns.

Related arguments: Pedersen (2009); delegated fund flows in Vayanos and Woolley (2012).

# Liquidity explains only part of the premium

| 48 global portfolios | Average $\lvert\alpha\rvert$ bp/month | $R^2_{\mathrm{CS}}$ |
|---|---|---|
| Global market | 35 | 0.52 |
| Market + liquidity factor | 31 | 0.54 |
| Market + global value + momentum | 18 | 0.72 |

Momentum's positive liquidity exposure may justify part of its premium. Value's negative exposure makes its positive premium harder to explain with this risk alone.

The 50/50 strategy retains strong returns with insignificant exposure to the global liquidity indices tested.

All three models fail the GRS test. Insignificance is not zero exposure: some U.S. liquidity proxies have significant combination loadings. Measurement error may also limit explanatory power.

# Conclusion

1. Value and momentum show a common structure across markets. Studying the styles jointly makes their diversification benefits visible.

2. Global value and momentum factors improve the description of returns, while leaving significant pricing errors.

3. Funding liquidity is a partial economic explanation. The size of the premia and their joint structure remain a challenge.

The contribution is a broader set of facts and a useful global benchmark.

# Thank you

Questions